Chapter 5
Orthogonality

5.3 Least Squares Problems

Motivation: Line of Best Fit

x	0	3	6	x
y	1	4	5	y

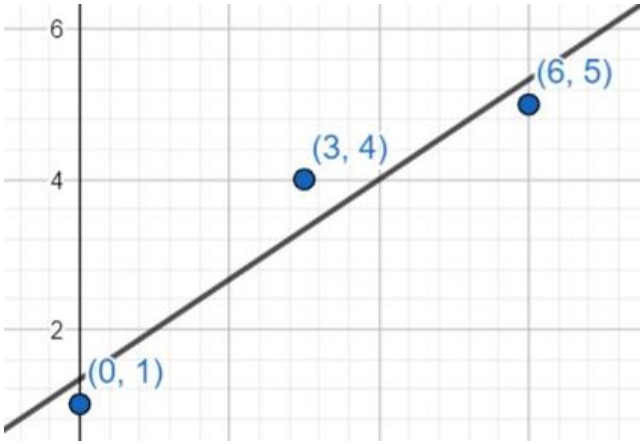
- Given a table of data,

we wish to find a linear function

$$y = c_0 + c_1 x$$

- From data, we get a linear system

$$\begin{bmatrix} 1 & 0 \\ 1 & 3 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$$



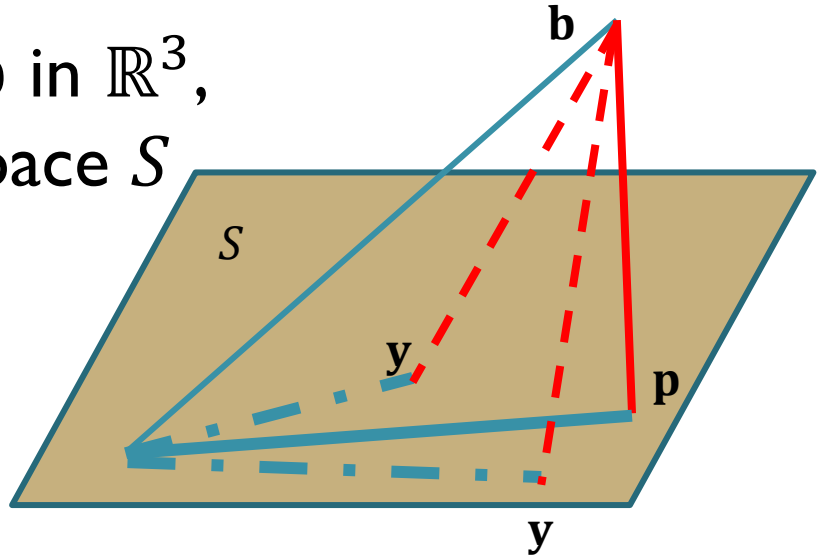
- This overdetermined system is inconsistent! $\mathbf{b} \notin R(A)$!
- No line passes through all 3 points!
- Each data point $[x_i, y_i]^T$ is d_i units above/below any line.
- **Question:** Can we still find a line that minimizes $d_1^2 + d_2^2 + d_3^2$?

Least Squares Solutions

- For an $m \times n$ matrix A, if the linear system $A\mathbf{x} = \mathbf{b}$ does not have a solution $\mathbf{x} \in \mathbb{R}^n$, we still want a vector $\hat{\mathbf{x}} \in \mathbb{R}^n$ such that $A\hat{\mathbf{x}}$ is closest to $\mathbf{b}$.
- If $r(\mathbf{x}) = \mathbf{b} - A\mathbf{x}$, we seek to minimize $\|r(\mathbf{x})\|$, which is the same as minimizing $\|r(\mathbf{x})\|^2$.
- A vector $\hat{\mathbf{x}} \in \mathbb{R}^n$ that minimizes $\|r(\mathbf{x})\|^2 = \|\mathbf{b} - A\mathbf{x}\|^2$ is called a least squares solution of $A\mathbf{x} = \mathbf{b}$.
- $\mathbf{p} = A\hat{\mathbf{x}}$ is the vector in $S = R(A)$, the column space of A, that is closest to $\mathbf{b}$.
- **We want to find $\mathbf{p} = A\hat{\mathbf{x}}$ that minimizes the distance between a point $\mathbf{b}$ and a subspace $S = R(A)$.**

Distance from a point to a plane

Suppose we have a vector $\mathbf{b}$ in $\mathbb{R}^3$,
$\mathbf{y}$ is any vector on the subspace S



The distance from point $\mathbf{b}$ to plane S can be defined as

$$d = \min_{\mathbf{y} \in S} \|\mathbf{b} - \mathbf{y}\|$$

Questions: is there a point on the plane that is closest to $\mathbf{b}$? How many closest points are there? How to find this point?

Exists Unique closest $\mathbf{p}$

Theorem Let S be a subspace of $\mathbb{R}^m$.

1. For each $\mathbf{b} \in \mathbb{R}^m$, there is a unique element $\mathbf{p}$ of S that is closest to $\mathbf{b}$:

$$\|\mathbf{b} - \mathbf{y}\| > \|\mathbf{b} - \mathbf{p}\|$$

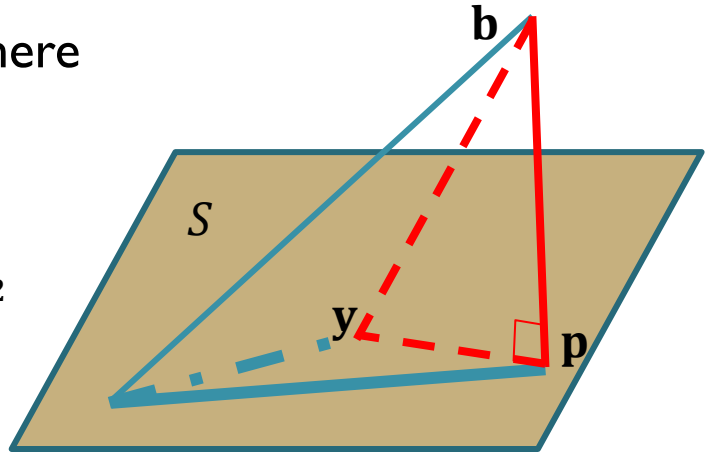
For any $\mathbf{y} \neq \mathbf{p}$ in S

2. A vector $\mathbf{p}$ in S is closest to a vector $\mathbf{b} \in \mathbb{R}^m$ if and only if $\mathbf{b} - \mathbf{p} \in S^{\perp}$.

Explanation

- [Finding $\mathbf{p}$:] $\mathbb{R}^m = S \oplus S^{\perp}$, so each $\mathbf{b} \in \mathbb{R}^m$ can be written uniquely as $\mathbf{b} = \mathbf{p} + \mathbf{z}$ where $\mathbf{p} \in S$ and $\mathbf{z} = \mathbf{b} - \mathbf{p} \in S^{\perp}$.
- [Closest distance and uniqueness:] By Pythagorean Law:

$$\|\mathbf{b} - \mathbf{y}\|^2 = \|\mathbf{b} - \mathbf{p}\|^2 + \|\mathbf{p} - \mathbf{y}\|^2 \geq \|\mathbf{b} - \mathbf{p}\|^2$$



Back to Least Squares Solution

- Least squares problem: subspace $S = R(A)$. $\mathbf{p} = A\hat{\mathbf{x}}$.
- $\mathbf{b} - \mathbf{p} = \mathbf{b} - A\hat{\mathbf{x}} = \mathbf{r}(\hat{\mathbf{x}}) \in R(A)^{\perp}$
- We have seen that $R(A)^{\perp} = N(A^T)$
- Therefore, $\mathbf{b} - A\hat{\mathbf{x}} \in N(A^T)$

$$\Rightarrow A^T(\mathbf{b} - A\hat{\mathbf{x}}) = A^T\mathbf{b} - A^TA\hat{\mathbf{x}} = \mathbf{0}$$

Summary: to solve the least squares problem

$$A\mathbf{x} = \mathbf{b}$$

normal equation

We need to solve

$$A^TA\mathbf{x} = A^T\mathbf{b}$$

- If A is $m \times n$, then A^TA is $n \times n$.

Normal Equation

Theorem

If A is an $m \times n$ matrix of rank n, the normal equation

$$A^T A\mathbf{x} = A^T\mathbf{b}$$

have a unique solution

$$\hat{\mathbf{x}} = \left(A^T A\right)^{-1} A^T\mathbf{b}$$

and $\hat{\mathbf{x}}$ is the unique least squares solution of $A\mathbf{x} = \mathbf{b}$.

Explanation

- A is $m \times n$ of rank $n \Rightarrow N(A) = \{\mathbf{0}\}$. So A has independent columns; in other words, columns of A is a basis of $R(A)$. (It also implies $m \geq n$, linear system is overdetermined.)
- By Exercise 5.2.13, $A^T A$ is nonsingular, so $\left(A^T A\right)^{-1}$ exists, and the normal equation has a unique solution.

Projection formula

If A is an $m \times n$ matrix of rank n (i.e. columns are independent),

- The unique least squares solution of $A\mathbf{x} = \mathbf{b}$ is

$$\hat{\mathbf{x}} = \left(A^T A\right)^{-1} A^T \mathbf{b}$$

- The projection of $\mathbf{b}$ onto $R(A)$ is

$$\mathbf{p} = A\hat{\mathbf{x}} = A\left(A^T A\right)^{-1} A^T \mathbf{b}$$

- Projection is a linear operator, with projection matrix

$$P = A\left(A^T A\right)^{-1} A^T$$

Example: $A = \mathbf{y}$ (matrix with 1 column), $R(A) = \mathrm{Span}(\mathbf{y})$,

$$\left(A^T A\right)^{-1} = \left(\mathbf{y}^T \mathbf{y}\right)^{-1} = \frac{1}{\|\mathbf{y}\|^2}; A\left(A^T A\right)^{-1} A^T \mathbf{b} = \frac{\mathbf{y}^T \mathbf{b}}{\|\mathbf{y}\|^2} \mathbf{y}$$

Which is exactly the vector projection of $\mathbf{b}$ onto $\mathbf{y}$!

Example 1: least squares solution

Find the least squares solution of the system

$$\begin{aligned} x_1 + x_2 &= 3 \\ -2x_1 + 3x_2 &= 1 \\ 2x_1 - x_2 &= 2 \end{aligned}$$

$$A\mathbf{x} \approx \mathbf{b}$$

$$\downarrow$$

$$A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$$

Solution

The normal equations for this system are

$$\begin{bmatrix} 1 & -2 & 2 \\ 1 & 3 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -2 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & -2 & 2 \\ 1 & 3 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 9 & -7 \\ -7 & 11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

Therefore, the least squares solution is

$$\hat{\mathbf{x}} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 9 & -7 \\ -7 & 11 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 83/50 \\ 71/50 \end{bmatrix}$$

$$\hat{\mathbf{x}} = \left(A^T A\right)^{-1} A^T\mathbf{b}$$

See next page for a follow-up

Example 1: projection

(Continued) The least squares solution of the system

$$\begin{aligned} x_1 + x_2 &= 3 \\ -2x_1 + 3x_2 &= 1 \\ 2x_1 - x_2 &= 2 \end{aligned} \qquad A\mathbf{x} \approx \mathbf{b}$$

is $\hat{\mathbf{x}} = \left(A^T A\right)^{-1} A^T \mathbf{b}$

$$\hat{\mathbf{x}} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 9 & -7 \\ -7 & 11 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 83/50 \\ 71/50 \end{bmatrix}$$

So the projection of $\mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$ onto the plane S spanned by $\begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$ is $\mathbf{p} = \begin{bmatrix} 1 & 1 \\ -2 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 83/50 \\ 71/50 \end{bmatrix} = \begin{bmatrix} 3.08 \\ 0.94 \\ 1.9 \end{bmatrix}$, very close to $\begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$!

$\mathbf{p} = A\hat{\mathbf{x}}$

Example 2

Find the least squares solutions to $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \\ 2 & -1 \\ 0 & 5 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 3 \end{bmatrix}.$$

Solution:

We write the normal equation $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$:

$$A^T A = \begin{bmatrix} 1 & 1 & 2 & 0 \\ 2 & 0 & -1 & 5 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 0 \\ 2 & -1 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 30 \end{bmatrix} \quad \Rightarrow (A^T A)^{-1} = \begin{bmatrix} 1/6 & 0 \\ 0 & 1/30 \end{bmatrix}$$

$$A^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 2 & 0 \\ 2 & 0 & -1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 18 \end{bmatrix}$$

Normal equation: $\begin{bmatrix} 6 & 0 \\ 0 & 30 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 18 \end{bmatrix}$

So $\hat{\mathbf{x}} = \begin{bmatrix} 6 & 0 \\ 0 & 30 \end{bmatrix}^{-1} \begin{bmatrix} -1 \\ 18 \end{bmatrix} = \begin{bmatrix} 1/6 & 0 \\ 0 & 1/30 \end{bmatrix} \begin{bmatrix} -1 \\ 18 \end{bmatrix} = \begin{bmatrix} -1/6 \\ 3/5 \end{bmatrix}$ is the least squares solution.

Application: Line of Best Fit

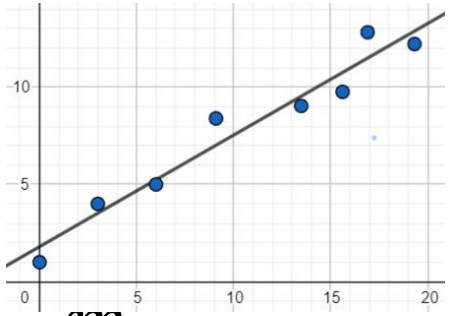
- Given a table of data

x	x_1	x_2	...	x_m
y	y_1	y_2	...	y_m

we wish to find a linear function

$$y = c_0 + c_1 x$$

that best fits the data.



- If we require that

$$y_i = c_0 + c_1 x_i \quad \text{for} \quad i = 1, \ldots, m$$

we get a system of m equations in 2 unknowns

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

- If c_0, c_1 are the least squares solutions, then $y = c_0 + c_1 x$ is called the <u>best least square fit by a linear function</u>.

Example: line of best fit

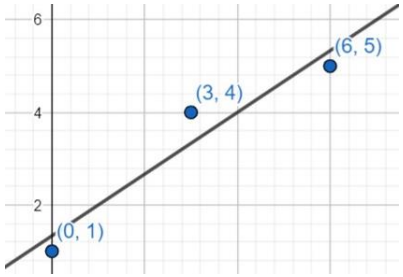
Given the data

x	0	3	6
y	1	4	5

Find the best least squares fit by a linear function.

Solution: We find the least squares solution to

$$\begin{bmatrix} 1 & 0 \\ 1 & 3 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}.$$



Normal equation:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 3 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 9 \\ 9 & 45 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} 10 \\ 42 \end{bmatrix} \quad \Rightarrow \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} 4/3 \\ 2/3 \end{bmatrix}$$

Therefore, the best least squares fit is $y = \frac{4}{3} + \frac{2}{3}x$.

Best Fit by a Polynomial

- Data table:

x	x_1	x_2	…	x_m
y	y_1	y_2	…	y_m

- Model with 3 parameters:

What about higher degree polynomials, or even other functions?

$$y = c_0 + c_1 x + c_2 x^2$$

- Each data point $[x_i, y_i]^T$ gives one equation:

$$y_i = c_0 + c_1 x_i + c_2 x_i^2 \quad \text{for} \quad i = 1, \ldots, m$$

- System of m equations in 3 unknowns:

$$\begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_1^2 \\ \vdots & \vdots & \vdots \\ 1 & x_m & x_m^2 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

- We can find the least squares solution $\hat{c}$, so that $y = c_0 + c_1 x + c_2 x^2$ will be the quadratic polynomial that gives the best least squares fit to the data.

Example: best quadratic fit

Given the data

x	0	1	**2**	3
y	3	2	4	4

Find the best least squares fit by $y = c_0 + c_1 x + c_2 x^2$.

Solution: We find the least squares solution to $A\mathbf{x} = \mathbf{b}$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 4 \\ 4 \end{bmatrix}.$$

Normal equation $A^T A\mathbf{x} = A^T \mathbf{b}$ (after simplification):

$$\begin{bmatrix} 4 & 6 & 14 \\ 6 & 14 & 36 \\ 14 & 36 & 98 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 13 \\ 22 \\ 54 \end{bmatrix}$$

$$\Rightarrow c_0 = 2.75, c_1 = -0.25, c_2 = 0.25$$

Therefore, the best least squares fit by quadratic functions is $y = 2.75 - 0.25x + 0.25x^2$.

Summary

- To solve the least squares problem

$$A\mathbf{x} = \mathbf{b}$$

We instead solve the normal equation

$$A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$$

- Projection of $\mathbf{b}$ onto $R(A)$ is

$$\mathbf{p} = A\hat{\mathbf{x}}$$

- Least squares solution $\hat{\mathbf{x}}$ minimizes

$$\|r(\mathbf{x})\|^2 = \|\mathbf{b} - A\mathbf{x}\|^2$$

and $\mathbf{b} - A\hat{\mathbf{x}}$ is orthogonal to $R(A)$.

About $A^T A$: preview of 5.5

- When calculating $A^T A$, we are doing scalar products of column vectors in A: $\left(A^T A\right)_{ij} = (\boldsymbol{a}_i)^T \boldsymbol{a}_j$
- $A^T A$ is symmetric; after calculating $(\boldsymbol{a}_2)^T \boldsymbol{a}_3$, no need to calculate $(\boldsymbol{a}_3)^T \boldsymbol{a}_2$ again!
- Normal equation $A^T A\mathbf{x} = A^T \mathbf{b}$ is the simplest to solve when $A^T A = I$; least squares solution $\mathbf{x} = A^T \mathbf{b}$.
- **QUESTION:** If $A^T A = I$, what do we know about the column vectors of A, $\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3$, ...? *This will be the motivation for the next part, 5.5*